

LAVISH GARG

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SUMMARY

Backend Developer with 3+ years of experience building scalable systems, integrating real-time AI solutions, and optimising performance through efficient backend design, SQL tuning, and automation.

SKILLS

- Languages:** JavaScript (Node.js), Python, PHP
- Frameworks & Architecture:** Node.js, Express.js, Yii2, Laravel, Monolithic, Microservices
- Databases & Storage:** MySQL, MongoDB, Redis, ClickHouse
- Streaming & Protocols:** REST, WebSocket, WebRTC, RTMP, HLS
- DevOps & Cloud:** AWS (EC2, S3, CloudFront), Docker, GitHub Actions, CI/CD Pipelines
- Performance & Observability:** SQL Tuning, Caching, Logging & Monitoring (Prometheus/Grafana)
- CS Fundamentals:** Data Structures & Algorithms, OOP, OS

EXPERIENCE

TEAM LEAD — VIDYAKUL

March 2026 – present

- Led backend architecture and development for large-scale test series, quiz, and student analytics infrastructure using Node.js, Redis queues, WebSockets, MongoDB, MySQL, and ClickHouse, serving 100K+ monthly active students and handling 50K+ daily test submissions during peak academic periods.
- Architected scalable real-time assessment systems including leaderboard computation, rank generation, auto-submission handling, timer synchronization, anti-cheating validations, and result processing pipelines, supporting 10K+ concurrent quiz participants with low latency response times.
- Built event-driven analytics services using in-house ClickHouse infrastructure for monitoring user engagement, API performance, live class metrics, and operational insights, processing 20M+ events daily.
- Developed advanced student intelligence dashboards featuring subject-wise weakness analysis, accuracy trends, performance benchmarking, engagement heatmaps, learning progression tracking, and personalized feedback insights used by 50K+ students and educators, improving reporting efficiency by 40%.

BACKEND DEVELOPER — VIDYAKUL

January 2025 – February 2026

- Engineered and integrated real-time AI voice models with Exotel Voice Streaming, automating outbound calls and reducing manual outreach by 40%, while improving operational scalability and team efficiency.
- Designed and built an in-house live video streaming platform using RTMP, HLS, Nginx-RTMP, and FFmpeg-based transcoding pipelines, enabling low-latency scalable delivery while reducing video streaming costs by 30%.
- Refactored and optimized complex SQL queries, indexing strategies, and backend workflows, reducing database CPU consumption by 16% and improving high-traffic API response performance.

FREELANCER — DELHI METRO RAIL CORPORATION

September 2024 – November 2024

- Developed a 96% accurate OCR-powered real-time monitoring system capable of extracting, validating, and storing critical alert messages from auto-scrolling operational event displays using live webcam feeds.
- Implemented multithreaded video-processing architecture capable of analyzing multiple frames per second with parallel OCR execution, image preprocessing, and high-speed data ingestion pipelines.
- Integrated real-time alert notifications, searchable logs, and downloadable reporting systems, reducing operational error resolution time by 30% and improving monitoring reliability and safety response workflows.

DEVELOPER — MYLO PREGNANCY & PARENTING

January 2023 – August 2024

- Managed centralized data management and analytics platforms used across 10+ internal departments for user tracking, order monitoring, P&L reporting, operational analytics, and Tally-integrated financial workflows.
- Implemented an in-house warehouse, inventory, and ledger management platform replacing third-party systems such as Unicommerce, reducing per-order operational commission costs by 2% while improving inventory visibility and fulfillment efficiency.
- Automated invoice generation, waybill processing, and logistics workflows, reducing manual paperwork by 90% and significantly improving dispatch efficiency
- Contributed to the development of an in-house distributor management platform used by 500+ distributor stores, replacing Bizom and improving order lifecycle management, inventory synchronization, distributor onboarding and operational efficiency.

EDUCATION

B.E. in Computer Science, Thapar Institute of Engineering & Technology, Patiala

August 2019 – Jun 2023